



Drug–target affinity prediction method based on multi-scale information interaction and graph optimization

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ABSTRACT

Drug–target affinity (DTA) prediction as an emerging and effective method is widely applied to explore the strength of drug–target interactions in drug development research. By predicting these interactions, researchers can assess the potential efficacy and safety of candidate drugs at an early stage, narrowing down the search space for therapeutic targets and accelerating the discovery and development of new drugs. However, existing DTA prediction models mainly use graphical representations of drug molecules, which lack information on interactions between individual substructures, thus affecting prediction accuracy and model interpretability. Therefore, transformer and diffusion on drug graphs in DTA prediction (TDGraphDTA) are introduced to predict drug–target interactions using multi-scale information interaction and graph optimization. An interactive module is integrated into feature extraction of drug and target features at different granularity levels. A diffusion model-based graph optimization module is proposed to improve the representation of molecular graph structures and enhance the interpretability of graph representations while obtaining optimal feature representations. In addition, TDGraphDTA improves the accuracy and reliability of predictions by capturing relationships and contextual information between molecular substructures. The performance of the proposed TDGraphDTA in DTA prediction was verified on three publicly available benchmark datasets (Davis, Metz, and KIBA). Compared with state-of-the-art baseline models, it achieved better results in terms of consistency index, R-squared, etc. Furthermore, compared with some existing methods, the proposed TDGraphDTA is demonstrated to have better structure capturing capabilities by visualizing the feature capturing capabilities of the model using Grad-AAM toxicity labels in the ToxCast dataset. The corresponding source codes are available at <https://github.com/Lamouryz/TDGraph>.

1. Introduction

Understanding the relationship between a drug and its target is crucial to drug discovery and development. Despite recent advances in biomedical research and technology, the field remains challenging. Predicting drug–target interactions (DTI) helps researchers evaluate drug candidates' potential effectiveness and safety early in the process, streamlining the search for disease-specific drugs [1]. Analyzing binding affinity is critical for assessing interactions between pharmaceutical compounds and target molecules such as receptors or protein kinases. By quantifying DTI binding affinity, the extent to which a drug binds

to a specific target can be reflected more effectively. Furthermore, quantifying the strength of these interactions provides more detailed and specific information than binary relationships [2].

In the early days, pharmacological and chemical methods were mainly used to predict drug–target interactions. Pharmacological methods involve analyzing the biological responses and properties of drugs and targets to predict interactions. For instance, methods such as structure–activity relationship (SAR) analysis based on drug action, drug metabolism kinetics, and gene expression profiling can be used to predict drug–target interactions [3]. Chemical methods focus on

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studying the molecular structures and physicochemical properties between drugs and targets to predict interactions, which include molecular docking simulations, quantitative structure–activity relationship (QSAR) models, and prediction models based on chemical structural similarity [4]. However, experimental measurements of drug–target interactions using pharmacological assays for millions of pharmaceutical compounds and hundreds of potential targets are time-consuming and resource-intensive [5]. Therefore, an efficient approach is needed to validate a large number of potential drug–target relationships.

With the continuous development of biotechnology and computer technology, machine learning methods have been applied to high-throughput computation-assisted experiments in new drug discovery, and have obtained good feedback on application effects [6]. Traditional machine learning methods such as support vector machine (SVM), random forest (RF), and artificial neural network (ANN) [7] have been widely used in QSAR, protein chemical similarity (PCM) methods [8] and molecular docking to simulate DTA [9]. Compared with pharmacological analysis methods, deep learning-based methods can automatically extract features from drugs and targets without the need for manual feature engineering. Due to their adaptability and ability to handle a variety of molecular structures, they are widely used in new drug development.

Typically, these machine learning-based or deep learning-based drug discovery modeling methods take drugs represented as one-dimensional sequences as input. However, this type of representation is insufficient to capture complex 2D/3D molecular structures, thus affecting the performance of predictive models. To solve this problem, graph neural networks have been used in drug discovery, including interaction prediction, molecular design [10], etc., which can handle irregular molecular structures through graphical representation.

Although the previously mentioned methods improve drug–target interaction prediction, increase accuracy and provide good results, some challenges still exist. Traditional graph neural networks have difficulty capturing long-range connections in molecular substructures and tend to focus on node information and ignore edge features. In DTA prediction, it is important to consider contextual information in the individual representations of proteins and targets and in the model structure, as well as important association information related to both entities. To address these problems, a DTA prediction based on multi-scale information interaction and graph optimization method is proposed. It integrates the idea of diffusion model into the optimization of drug molecular graphs, combining graph diffusion and sparsification to enhance feature representation. In addition, a multi-scale interaction block is proposed to efficiently integrate multi-scale information features of drugs and targets to capture the relationships and contextual information between molecular substructures. The joint representation of drug–target pairs improves prediction accuracy and reliability.

This paper has three main contributions as follows.

- The TDGraphDTA model is proposed to solve the optimal representation of graphs and the fusion of multi-scale features in DTA prediction. It consists of an interaction block for feature extraction from drugs and targets at different scales and a graph optimization block based on a diffusion model for optimizing the molecular graph structure.
- According to the idea of diffusion model, the representation of relevant information between nodes and the graph structure are optimized during the construction process of the molecular graph, and additional molecular structure features are explored to enrich feature information. Additionally, the theoretical interpretability of the model is improved.
- A multi-scale interaction block is proposed. According to the multi-head self-attention idea of Transformer, the interactions of drug and target information at different granularities are integrated. Important features associated with two entities can be

enhanced by capturing the relationships and contextual information between molecular substructures. Through the joint representation of drug–target pairs, both the accuracy and reliability of predictions are improved.

The remaining sections of this paper are organized as follows. Section 2 provides an overview of the relevant literature on deep learning-based DTA prediction. Section 3 specifies the three modules of the proposed model, input module, molecular representation optimization learning module, and output module. Section 4 discusses the experimental datasets, settings, and evaluations of the proposed method on three benchmark datasets (i.e., Davis, Metz, and KIBA [11]), and the corresponding results and analysis. Section 5 concludes this paper and points out future research directions.

2. Related work

Deep learning methods are now widely applied to various research fields, including bioinformatics (e.g., genomic studies) [12], and QSAR studies in drug discovery. As a major advantage, deep learning methods can well represent the raw data through nonlinear transformations at each layer [13], thereby facilitating the discovery of hidden patterns. These deep learning-based methods have broad applications in predicting new drug mechanisms and target binding sites, leading to improved prediction accuracy and reliability. According to the form of the drug–target input to the neural networks, they can be roughly classified into three categories, sequence-based methods for drugs and targets, graph-based methods for drug molecules, and methods based on the three-dimensional structure of proteins.

2.1. Drug–target affinity prediction method based on drug and target sequences

A method for protein sequence and composite one-dimensional representations using convolutional neural networks (CNNs) (short for DeepDTA) proposed by Öztürk et al. [1] first established separate CNNs to learn representations of drugs and proteins. These representations were then connected and fed into a multilayer perceptron (MLP) for DTA prediction. As hybrid neural networks, a new efficient hybrid neural network model to predict DTI (DTI-RCNN) was introduced by Zheng et al. [14] based on the above-mentioned approach, which integrated recursive neural networks into CNNs. A method for predicting DTI via deep learning and protein sequence convolution (DeepConv-DTI) developed by Lee et al. [15] uses amino acid sequences and drug fingerprint strings as input. Local residue conformation patterns in proteins are effectively captured using one-dimensional CNNs (1D-CNNs) with various-sized convolutional kernels, reducing the risks associated with inappropriate kernel sizes. Inspired by DeepDTA, a method for deep cross-domain compound–protein affinity prediction via long short-term memory (LSTM) and CNNs (DeepCDA) proposed by Abbasi et al. [16] incorporates LSTM layers after each CNN layer to enhance the model's ability to capture biological sequence information. DeepCDA clarifies the importance of molecular interaction information and predicts the binding regions of interactions in the model with a weight matrix of attention mechanism. A deep-learning based prediction model that employs chemical and biological textual sequence information to predict binding affinity (WideDTA) proposed by Öztürk et al. [17] integrates two additional text-based inputs and shares structure and protein motifs and domains to improve model performance. Four CNNs are employed to encode these inputs into four representations, further enhancing model performance. A method for predicting DTA using attention models presented by Zhao et al. [18] uses two independent 1D-CNNs to extract semantic information from drug SMILES strings and protein amino acid sequences, aiming to explore the relationship between drug and protein features.

2.2. Drug–target affinity prediction method based on drug molecular map

Tsubaki et al. [19] developed a novel DTI prediction method that integrates the graph neural networks (GNNs) for drugs, the CNNs for proteins, and an attention mechanism used to integrate representations. The accuracy of DTI prediction models can be further enhanced by using molecular graphs to represent drug features [20]. As a large-scale DTI prediction system, a method for high-performance DTI prediction via CNNs using two-dimensional structural compound representation (DEEPScreen) proposed by Rifaioğlu et al. [21] represents compounds as 2D structure images and uses CNN to learn complex features from these images, resulting in high-precision DTA prediction. Nguyen et al. [22] introduced a graph neural network-based approach (GraphDTA) to predict the binding affinity between drugs and targets. In GraphDTA, drugs are represented as graphs and GNNs are used to predict DTA. Building upon this, Cheng et al. [23] proposed an end-to-end deep learning DTA prediction method based on graph attention networks and multi-head self-attention mechanism. A method for predicting DTA using GNNs and contact graphs (DGraphDTA) proposed by Jiang et al. [24] utilizes the structural information of molecules and proteins to construct separate graphs for drug molecules and proteins, and employs GNNs on both sides to obtain their representations. Yang et al. [11] introduced deep multiscale GNNs for interpretable DTA prediction (MGraphDTA) to incorporate dense connections into GNNs and construct super-deep GNNs with 27 graph convolutional layers to simultaneously capture the local and global structures of compounds. The novel graph-in-graph neural networks with attention mechanism (GEFA) were proposed by Nguyen et al. [25] to address target representation variations caused by binding effects. Liao et al. [26] introduced an interpretable deep learning framework, which integrates a self-attention mechanism and GNNs to construct representations of drugs and target proteins from structural information. They also introduced mutual information to filter out redundant information and retain relevant information in the combination of drugs and targets.

2.3. Drug–target affinity prediction method based on the 3D structure of protein

Wallach et al. [27] introduced a pioneering deep learning model AtomNet, in which three-dimensional convolutional neural networks (3D-CNNs) were used to process the three-dimensional structure of drug–target complexes. Gomes et al. [28] proposed the atomic convolutional neural networks (ACNNs) to predict binding affinity. ACNNs extract the local features of the molecular and biochemical environment in the encoded information through radial pooling convolution operations, which reduce the complexity of model calculation. Jiménez et al. [29] presented an end-to-end framework based on 3D-CNNs for accurate prediction of absolute binding affinity in drug–target pairs. Considering the computational complexity of long molecular sequences, DeepAtom as the efficient and lightweight neural networks based on 3D-CNNs was proposed by Li et al. [30]. The networks automatically capture atomic interaction patterns related to binding affinities, improving computational efficiency while ensuring certain predictive performance. Lim et al. [31] introduced a distance-aware graph attention algorithm to distinguish various types of intermolecular interactions, extracting graph features of molecular interactions directly from the 3D structural information of protein–ligand binding poses. Voitsitskyi et al. [32] developed a novel deep learning DTA model that incorporated AlphaFold structure predictions and protein graph representations.

Most sequence-based DTA prediction methods represent drugs as strings, which is not a natural way to represent chemical compounds. The use of strings results in a loss of molecular structural information, which may affect the ability of predictive models and the internal learning space of functional correlations. 3D representation of drugs

can contain richer structural information, but availability to quantify intermolecular affinities often relies on experimentally determined co-crystal structures [33]. However, most drug–target pairs usually lack such structures, which also makes the molecular 3D structure affinity characterization data difficult to use in large quantities. Furthermore, obtaining the structures of small molecules and proteins in 3D is challenging and computationally complex. These restrictions limit applicability of 3D structure-based methods. Therefore, an innovative prediction method is proposed to combine graph optimization with the interaction of drug–target substructure features. By exploiting the internal structure of the architecture, the proposed method can achieve optimal representation of drug–target features and potential interpretability for highly accurate predictions.

3. The proposed method

Fig. 1 shows the overall framework of the proposed DTA prediction solution, which integrates multi-scale information interaction and graph optimization. It consists of three main modules, the input module, the molecular representation optimization learning module, and the output module. In the input module, protein sequences and drug graphs built using RDKit [34] are used as network input. The molecular representation optimization learning module extracts features from protein sequences and molecular graphs separately, and fuses them into respective representations. These representations are then fed into the multi-scale interaction block. The output module combines feature maps obtained from the multi-scale interaction block, containing weighted and substructure interaction information. These feature maps are connected and fed into fully connected layers to predict affinity. This section elaborates the proposed Graph optimization block based on the diffusion model and the multi-scale interaction block.

3.1. Input module

The whole model includes two inputs, as illustrated in Fig. 2, namely the protein and drug input representations. In this module, different preprocessing transformations are performed on drug and protein sequences, respectively. Drug sequences are converted into computer-friendly one-dimensional linear numerical symbols. Protein sequences are converted into a computer data structure of a graph consisting of vertices and edges. The drug input representation is expressed as simplified molecular input line entry system (SMILES) [1], a linear symbol for the input and representation of molecular reactions. SMILES employs a concise ASCII string to describe the structure of chemical compounds. Instead, protein input consists of amino acid sequences, where each character represents a specific amino acid. Regarding the drug's SMILES input, it is preprocessed using the RDkit tool [11] to generate a graph with node features and an adjacency matrix. This graph structure shows molecules as collections of atoms forming nodes, with bonds as edges. It effectively captures the connectivity between atoms, including nodes and edge features, as shown in Fig. 2(b).

3.2. Molecular representation optimization learning module

The molecular representation optimization learning module consists of protein sequence feature representation learning, compound molecular graph feature representation learning, and multi-scale interaction block.

3.2.1. Protein sequence feature representation learning

The protein multi-scale feature extraction networks are designed to extract multi-scale features from proteins. These branches consist of convolutional layers with different depths. By stacking the convolutional layers and sequentially passing their outputs through max-pooling layers, the network can capture local residue representations of proteins at different scales. For an input protein sequence of length

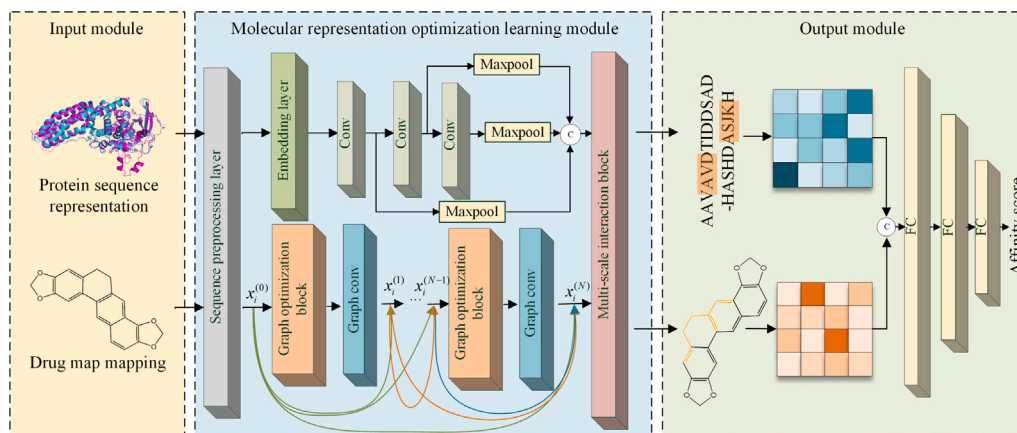


Fig. 1. The overall framework of the proposed method. The proposed method consists of three modules. The input module handles protein and drug inputs in various forms. The molecular representation optimization learning module performs the extraction and fusion of multi-dimensional features from drugs and protein targets. The output module combines the optimal feature representations of drugs and protein targets to predict affinity.

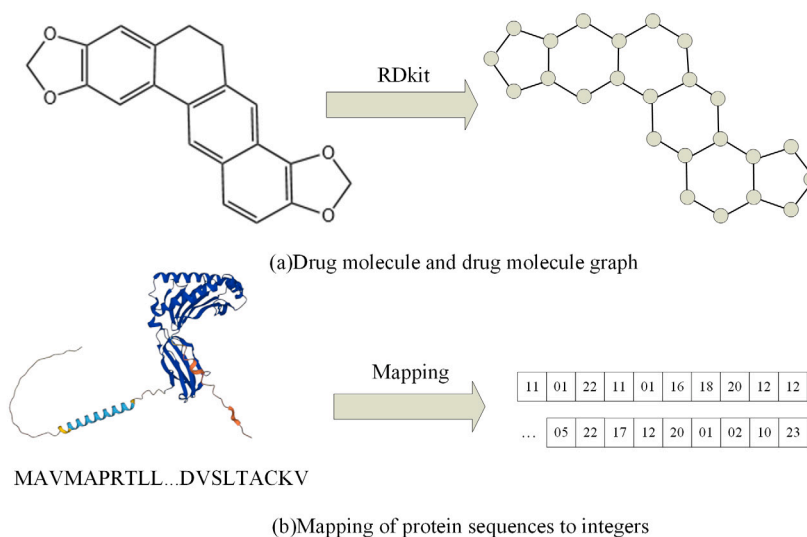


Fig. 2. Different characterization modalities for drug and protein targets. (a) Drug molecule graph representation and (b) Protein sequence representation.

1200 represented as integers, an embedding layer is first used to map each integer to a 128-dimensional vector, resulting in an input matrix $s \in \mathbb{R}^{1200 \times 128}$. This matrix is then fed into the protein feature extraction network to obtain the protein's feature representation Y_p .

$$Y_p = R \sum_{f=1}^3 \text{Maxpool}(C_k(C_{k-1} \dots (C_k(s)))) \quad (1)$$

where R denotes reshape to a tensor with the same size, Maxpool represents the maxpool operation mapped to an h -dimensional vector, C denotes a standard convolutional layer, $y_s \in \mathbb{R}^d$, f denotes a convolutional branch, and k represents the number of standard convolutional layers in each branch f .

3.2.2. Compound molecular graph feature representation learning

First, the constructed molecular graph of the drug is denoted as $G = (v, \epsilon)$, where v represents the set of vertices and ϵ represents the set of edges. In a molecule, v_i corresponds to the i th atom, and e_{ij} represents the chemical bond between the i th and j th atoms. After representing the drug compound as a graph, a diffusion-based Graph optimization block is designed to extract graph representation features. The resulting graph representation is then fed into graph convolution to obtain more meaningful and interconnected feature representations.

This graph diffusion form is defined by the following equation.

$$\mathbb{D}_{\text{graph}} = \frac{1}{L} \sum_{l=0}^L \sum_{g=0}^G \gamma_g M^g \quad (2)$$

where l/L denotes the number of diffusion convolution layers, ranging from l to L . g/G denotes the number of nodes of the graph, ranging from g to G . $\gamma_g \in (0 \sim 1)$ denotes the weighting factor and M^g denotes the transfer matrix of the corresponding graph.

To optimize graph learning methods, diffusion theory principles are applied [35], introducing noise into the system by simulating the random movement of particles or proxies within the networks or graph. By capturing the random correlations between nodes, this process adjusts and optimizes the weight distribution of nodes and edges, enhancing the expression of feature relationships and interpretability of the molecular graph. Specifically, this process adopts the form of sparse generalized graph diffusion, optimizing the constructed molecular graph through the process of diffusion and sparsification, as illustrated in Fig. 3.

It introduces noise into the system (i.e., shown in Eq. (3)), which is crucial for capturing the stochastic nature of partial interactions between drugs and proteins.

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} z_t \quad (3)$$

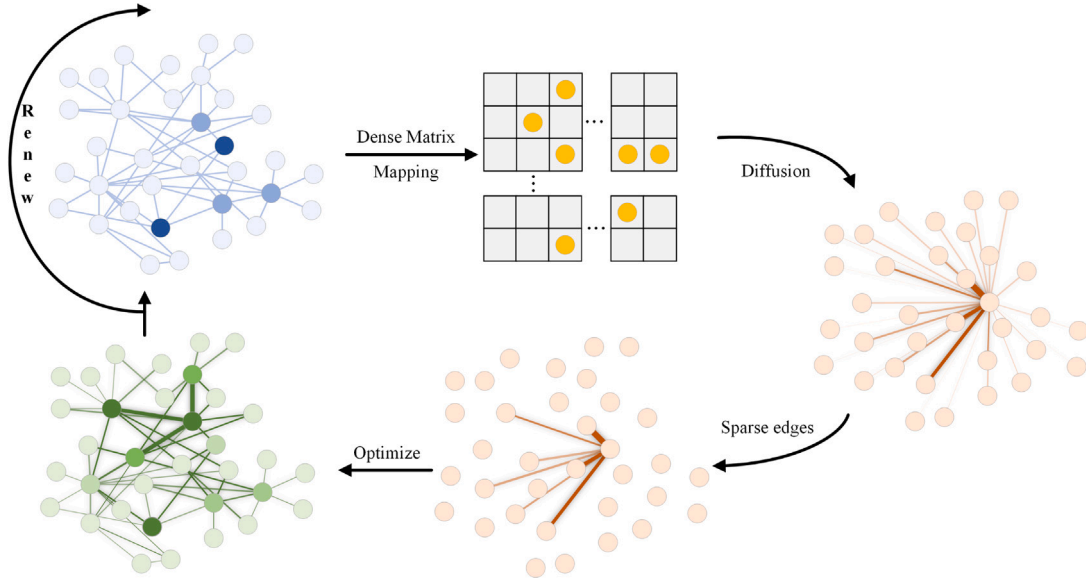


Fig. 3. The diffusion-based graph optimization process. The original graph is continuously optimized and reconstructed through graph diffusion and sparsification.

where x_t represents the information distribution at time t , z_1 denotes noise following a Gaussian distribution, and α_t represents the noise weight at time t .

Initially, The heat kernel [36] diffusion is used to define the information propagation paths on the graph, and the corresponding diffusion matrix is constructed to capture the pairwise similarity between nodes in the graph. The diffusion matrix is defined as the matrix exponential of a scaled version of the graph Laplacian operator, encoding pairwise similarities between nodes and incorporating higher-order interactions through the matrix exponential of the Laplacian matrix. The Laplacian matrix is defined as the difference between the degree matrix and the adjacency matrix of the graph as follows.

$$L = D - A \quad (4)$$

where D is the diagonal matrix of node degrees and $A \in \mathbb{R}^{|G| \times |G|}$ is the adjacency matrix and L is the Laplacian matrix. The construction of the diffusion matrix is given as follows.

$$\text{diff}_{\text{matrix}} = e^{(-\alpha L)} \quad (5)$$

where $\alpha \in (0 \sim 1)$ is the scaling parameter and e represents a mathematical constant.

A diffusion matrix is constructed to smooth the features of each node in the graph. After the signals propagate through the graph using a diffusion matrix, the smoothed signals capture both local and global structure, representing the probability of particles moving from one node to another. The signal smoothing is computed as follows.

$$\text{smoothed}_x = \text{diff}_{\text{matrix}} \cdot x_t \quad (6)$$

where x represents the vector of node features. The smoothed signals are subsequently utilized as input for the trained networks that predict the target variables. Training is conducted using backpropagation and stochastic gradient descent to minimize the loss function, which measures the disparity between predicted and actual values. By learning from smoothed signals, the networks capture the underlying structure of the graph, enabling accurate predictions.

Considering the randomness of diffusion in undirected graphs with row/column eigenvalues, a weighted self-loop is added to the original adjacency matrix to adjust the tensor random wandering. The self-loop edges are recorded and the least weighted of the parallel edges is retained to eliminate the parallel edges and avoid the invalid duplicate superposition of weights. Then the transformation matrix can be

denoted as the following equation.

$$T = (\Delta_{\text{loop}} \cdot \text{smoothed}_x + A)(\Delta_{\text{loop}} \cdot \text{smoothed}_x + D)^{-1} \quad (7)$$

where $\Delta_{\text{loop}} \in \mathbb{R}^+$ denotes the self-loop factor. Then the graph convolution diffusion form can be denoted as the following equation.

$$\mathbb{D}_{\text{graph}} = \frac{1}{l} \sum_{l=0}^L \sum_{g=0}^G (\Delta_{\text{loop}} \cdot \text{smoothed}_x + A)^G \quad (8)$$

$$\frac{1}{l} \sum_{l=0}^L \sum_{g=0}^G (\Delta_{\text{loop}} \cdot \text{smoothed}_x + D)^{-G}$$

3.2.3. Multi-scale interaction block

To capture the relationships and contextual information between molecular substructures and enhance the relevant features, a cross-attention mechanism is applied to handle the multimodal inputs of proteins and ligands, obtaining a joint representation of drug–target pairs. Specifically, the feature vectors of proteins and ligands are treated as two input sequences and the cross-attention mechanism is used to model their interaction. The specific network structure of this block is illustrated in Fig. 4.

In the cross-attention mechanism, the attention weights between two input vectors are first calculated. The protein feature vector is represented as X , while the drug molecule feature vector is represented as Y . For each drug subgraph, the cross-attention score matrix is calculated to measure its similarity with all target subgraphs.

The computation of the cross-attention score matrix follows a similar approach to the standard attention mechanism. It utilizes a multi-head attention mechanism to perform weighted summation. The computation process is outlined as follows.

Using the multi-head attention mechanism, X and Y are transformed into the query matrix Q , key matrix K , and value matrix V . Specifically, X and Y are linearly transformed to obtain query matrices Q_X and Q_Y , key matrices K_X and K_Y , and value matrices V_X and V_Y , where H represents the number of attention heads, d_k and d_v represent the dimensions of the key and value feature vectors, respectively.

To compute the attention scores, a dot product between the query matrix Q_X and the key matrix K_Y is performed, resulting in an attention score matrix S . S is then normalized using the softmax function, yielding a probability distribution matrix A . Next, a weighted summation of A and V_Y is performed to get a weighted matrix C_X . Similarly, a

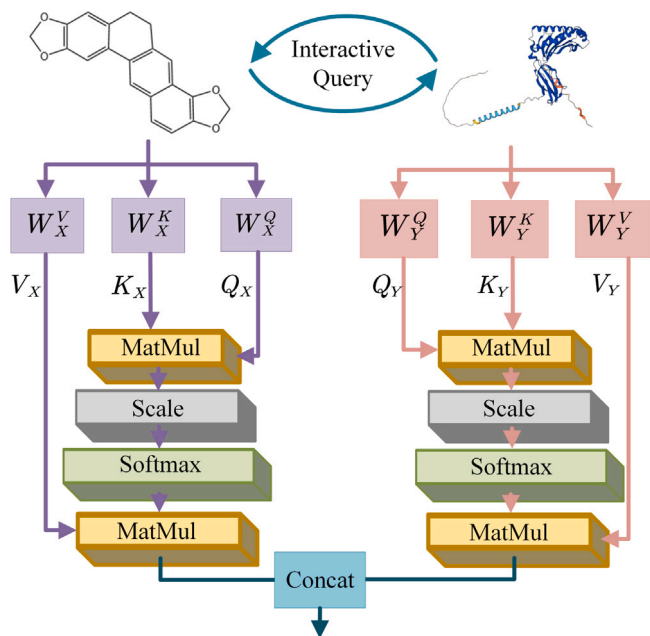


Fig. 4. The multi-scale interaction block. The outputs of the last two parallel cross-attention layers in the multi-scale interaction block are combined to obtain the weights representing the local and global correlations of the interaction. Among them, the MatMul layer describes the linear relationship between them by combining different matrix information, the Scale layer adjusts the weight of the feature map to match the context output shape, and the Softmax layer is used for the conversion of the feature vector and the probability distribution.

weighted summation of A and V_X is performed to get a weighted matrix C_Y .

C_X and C_Y are concatenated to obtain a matrix C , which represents the cross-attention representation of each element in X with each element in Y across multiple attention heads. The cross-attention mechanism is defined as follows.

$$CrossAtt_{protein} = \text{soft max} \left(\frac{Q_Y K_X^T}{\sqrt{d_{k_X}}} M_p \right) V_X \quad (9)$$

$$CrossAtt_{drug} = \text{soft max} \left(\frac{Q_X K_Y^T}{\sqrt{d_{k_Y}}} M_d \right) V_Y \quad (10)$$

The query matrix Q_X is computed from X , while the key matrix K_X and value matrix V_X are computed from Y . However, the cross-attention mechanism for compound molecules has the opposite input. The parameters W_X^K , W_X^V , W_X^Q , W_Y^K , W_Y^V and W_Y^Q are learnable, d_{k_X} and d_{k_Y} represent the dimensions of matrices K_X and K_Y , respectively. The attention score matrix $\text{soft max}(\frac{Q_Y K_X^T}{\sqrt{d_{k_X}}})$ or $\text{soft max}(\frac{Q_X K_Y^T}{\sqrt{d_{k_Y}}})$ repre-

sents the similarity between each amino acid in the pocket sequence and each atom in the ligand string. M_d/M_p denoted as the corresponding weight matrix. A multi-head cross-attention mechanism with two attention heads is utilized, and the outputs of the two parallel cross-attention layers are concatenated to obtain the local/global correlation weights for interaction.

3.3. Output module

Once the vector representations of the drug and target, and the local/global correlation weights are obtained, these two representations are concatenated and passed through a fully connected layer to predict

Table 1

Summary of the datasets.

Dataset	Compounds	Proteins	Number of paired samples
Davis	68	442	30,056
Metz	1423	170	35,259
KIBA	2111	229	118,254

the binding affinity score. By using the new tensor as a connection bridge of different feature map matrices, feature map matrices with different feature expression data are spliced horizontally according to the specified matrix connection dimension and data type. More specifically, the output module comprises three linear transformation layers that map the combined representation to an affinity score. Each linear transformation layer is followed by a ReLU activation function and a dropout layer. The activation function is given as follows.

$$ReLu(o) = \max(o, 0) = \begin{cases} o, & o > 0 \\ 0, & o < 0 \end{cases} \quad (11)$$

where o represents the input feature tensor.

After three conversions of the above neural network layers, the input feature size is converted from 1024 to 256 with the dropout parameter set to 0.1. Finally, the feature tensor goes through a linear layer with an input feature size of 256 to output the probability. The whole output module can be expressed as the following formula.

$$y_{pre} = f(\text{Linear}(\text{Relu}(xW^T + \zeta) \odot m)) \quad (12)$$

where W^T represents the transpose of the weight matrix of the layer, x denotes the input feature tensor, ζ is the bias term, $m \sim \text{Bernoulli}(u)$ is the layer mask and follows the Bernoulli distribution, u is the dropout probability (Each element in the mask has probability u), and f represents the forward propagation process.

4. Experiment

4.1. Dataset and implementation details

The proposed model is evaluated on three different datasets, Davis, Metz, and KIBA [11]. The binding affinities in these datasets are represented by dissociation constants (K_d), inhibition constants (K_i), and KIBA scores, which indicate the strength of binding between proteins and compounds. In the Davis and Metz datasets, the binding affinities increase with higher values of K_d and K_i , while the binding affinities decrease with increasing KIBA scores in the KIBA dataset. For calculation convenience, K_d and K_i are transformed into pK_d and pK_i , respectively. The Davis dataset comprises interactions between 442 proteins and 68 compounds, the Metz dataset includes interactions between 1423 proteins and 170 compounds, and the KIBA dataset consists of interactions between 2111 proteins and 229 compounds. Table 1 provides a summary of these datasets. Fig. 5 presents experimental summaries of the binding affinity distribution, SMILES string length distribution, and protein sequence length distribution in the Davis, Metz, and KIBA datasets. All data sets are divided into training set and test set according to the total number of drug-target pairs in the ratio of 8 : 2.

All experiments and field tests were conducted on a desktop computer with the following specifications, NVIDIA GeForce GTX 3090 with 24.00 GB of memory, Intel Core i7-8700MQ CPU @ 3.20 GHz, and 48.00 GB of RAM. Five-fold cross-validation were performed and the Optuna [37] automatic hyperparameter optimization software was employed to optimize the model hyperparameters on the Davis training set. The optimized hyperparameters were then used for all other datasets. The detailed hyperparameter settings are presented in [37] Table 2. Specifically, the protein and drug encoding layers were chosen from [2, 4, 8] and [4, 6, 8], respectively. The target sequence filter sizes

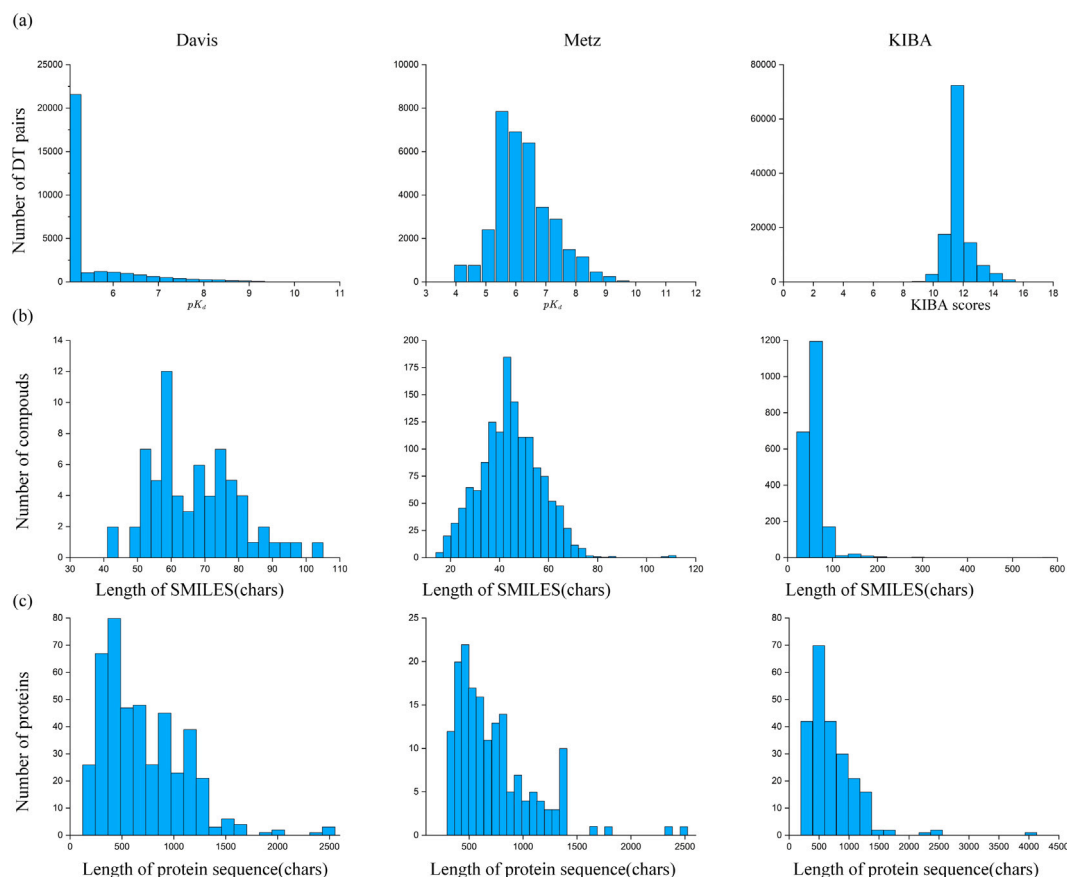


Fig. 5. Distribution of the Davis, Metz, and KIBA datasets. (A) Combining the distribution of affinity values. (B) Distribution of SMILES string lengths. (C) Distribution of protein sequence lengths.

Table 2
Experimental parameters.

Parameters	Range
Protein encoder layer	[2, 4, 8]
Drug encoder layer	[4, 6, 8]
Target Sequences strings filters length	(4, 3, 2)
Classifier layer	(32, 3, 2)
Learning rate	5e-4
Batch size	1024
Epoch	3000
The rate of drop out	1e-1
Regularization parameters	[1e-3, 5e-3, 7e-3]

were selected from (4, 3, 2). A batch size of 1024 and a regularization parameter λ chosen from [1e-3, 5e-3, 7e-3] were used, and training was performed for 3000 epochs. The Adam optimizer [32] with a learning rate of 0.0005 was used to update model parameters.

4.2. Performance comparison with different drug–target affinity prediction models

To verify the effectiveness of the proposed TDGraphDTA model, the predictive performance of previous methods is evaluated on the Davis, Metz, and KIBA datasets. Three quantitative metrics as performance indicators are applied to regression-based models, mean squared error (i.e., MSE, a smaller value means better performance), concordance index (i.e., CI, a larger value means better performance) [38], and corrected squared correlation coefficient (r^2 , a larger value means better performance) [39]. The proposed TDGraphDTA model is compared with traditional PCM models [40–43] (RF, SVM, FNN, Kron-RLS, and SimBoost), as well as state-of-the-art models (DeepDTA [1],

ML-DTI [44], WideDTA [17], DeepCDA [16], AttentionDTA [18], FusionDTA [45], GraphDTA [22], and MGraphDTA [11]).

For the regression task on the Davis and KIBA datasets, the proposed TDGraphDTA was compared with state-of-the-art methods, namely MGraphDTA and DeepDTA. The entire dataset was randomly split into six parts. Five parts were used for cross-validation and the remaining part was served as an independent testing dataset.

The final performance was evaluated on the independent testing dataset using the same setup of MGraphDTA. To ensure a fair comparison, the data points in each fold were consistent with the data points used in MGraphDTA.

Tables 3 and 4 demonstrate the performance of the TDGraphDTA method compared with other methods on the Davis and KIBA datasets. In both datasets, TDGraphDTA outperformed traditional machine learning methods and structure-based methods such as RF, SVM, Kron-RLS, and DeepDTA, establishing its comprehensive superiority. It also achieved favorable performance compared with sequence-based learning methods like DeepCDA with LSTM and ML-DTI with attention mechanism, indicating the rationality and effectiveness of the proposed multi-scale interaction block. When considering DTA prediction methods that used molecular graphs as model input, such as GraphDTA and MGraphDTA, TDGraphDTA performed best in terms of three metrics on the Davis dataset and approached the best performance in terms of CI on the KIBA dataset. This highlighted the contributions of graph optimization and information interaction in the regression process. Moreover, TDGraphDTA outperformed in terms of MSE and r^2 compared with other methods on both the Davis and KIBA datasets, showcasing the stability and generalization capability of the model structure. Overall, Tables 3 and 4 confirm that TDGraphDTA consistently achieved superior performance across the three metrics on the Davis and KIBA datasets compared with other methods.

Table 3

Comparison of the performance of TDGraphDTA with previous classical models on the Davis dataset. The best results are marked in bold.

Dataset	Category		Davis		
	Proteins	Compounds	MSE	CI(std)	r^2
RF	ECFP	PSC	0.359	0.854(0.002)	0.549(0.005)
SVM	ECFP	PSC	0.383	0.857(0.001)	0.513(0.003)
FNN	ECFP	PSC	0.244	0.893(0.003)	0.685(0.015)
KronRLS	S-W	P-S	0.379	0.871(0.0008)	0.407(0.005)
SimBoost	S-W	P-S	0.282	0.872(0.002)	0.644(0.006)
DeepDTA	CNN	P-S	0.419	0.835(0.005)	–
DeepDTA	S-W	CNN	0.420	0.886(0.008)	0.557(0.047)
DeepDTA	CNN	CNN	0.261	0.878(0.004)	0.630(0.017)
ML-DTI	CNN + Attention	CNN + Attention	0.197	0.886(0.009)	0.728(0.017)
WideDTA	PDM	LMCS	0.276	0.883(0.003)	–
DeepCDA	LSTM	LSTM	0.287	0.864(0.002)	0.620(0.021)
AttentionDTA	Transformer + CNN	Transformer + CNN	0.216	0.893(0.005)	0.677(0.024)
FusionDTA	Transformer + BiLSTM	BiLSTM	0.220	0.903(0.002)	0.666(0.008)
GraphDTA	CNN	GCN	0.204	0.883(0.008)	0.726(0.028)
MGraphDTA	MCNN	MGNN	0.207	0.900(0.004)	0.710(0.005)
TDGraphDTA	MCNN	With Transformer	0.212	0.904(0.005)	0.732(0.009)
TDGraphDTA	MCNN	With Diffusion	0.226	0.901(0.007)	0.721(0.013)
TDGraphDTA	MCNN	MD-GNN	0.199	0.906(0.003)	0.742(0.004)

Table 4

Comparison of the performance of TDGraphDTA with previous classical models on the KIBA dataset. The best results are marked in bold.

Dataset	Category		KIBA		
	Proteins	Compounds	MSE	CI(std)	r^2
RF	ECFP	PSC	0.245	0.837(0.000)	0.581(0.000)
SVM	ECFP	PSC	0.308	0.799(0.001)	0.513(0.004)
FNN	ECFP	PSC	0.216	0.818(0.005)	0.659(0.015)
KronRLS	S-W	P-S	0.411	0.782(0.001)	0.342(0.001)
SimBoost	S-W	P-S	0.222	0.836(0.001)	0.629(0.007)
DeepDTA	CNN	P-S	0.571	0.718(0.004)	–
DeepDTA	S-W	CNN	0.204	0.854(0.001)	0.692(0.009)
DeepDTA	CNN	CNN	0.194	0.863(0.002)	0.673(0.009)
ML-DTI	CNN + Attention	CNN + Attention	0.196	0.862(0.006)	0.727(0.012)
WideDTA	PDM	LMCS	0.179	0.875(0.001)	–
DeepCDA	LSTM	LSTM	0.214	0.852(0.012)	0.633(0.024)
AttentionDTA	Transformer + CNN	Transformer + CNN	0.214	0.852(0.012)	0.633(0.024)
FusionDTA	Transformer + BiLSTM	BiLSTM	0.155	0.882(0.004)	0.755(0.017)
GraphDTA	CNN	GCN	0.167	0.890(0.001)	0.699(0.010)
MGraphDTA	MCNN	MGNN	0.128	0.902(0.001)	0.801(0.001)
TDGraphDTA	MCNN	With Transformer	0.124	0.895(0.005)	0.810(0.001)
TDGraphDTA	MCNN	With Diffusion	0.145	0.889(0.002)	0.804(0.003)
TDGraphDTA	MCNN	MD-GNN	0.121	0.899(0.003)	0.817(0.003)

To further evaluate the model's performance in DTA prediction with molecular graph inputs, the Metz dataset was randomly split into training and testing sets. Parts of the models used for performance comparison were replaced on the Davis and KIBA datasets with different graph-based compound molecular graph learning methods. The results showed that TDGraphDTA performed well in terms of MSE, CI, and r^2 , with particularly strong performance in terms of CI and r^2 . Graph-based methods exhibited more advantages compared with CNN-based and RNN-based approaches, highlighting the potential of GNNs in DTA prediction. While CNN-based and RNN-based models represented compounds as strings, ignoring molecular structural information may impair their predictive power. In contrast, graph-based methods represented compounds as graphs and captured their dependencies by propagating messages between vertices. Compared with other graph-based methods, the proposed TDGraphDTA approach demonstrated the best overall performance across all three metrics.

Overall, the proposed TDGraphDTA method showed impressive results on the three benchmark datasets, Davis and Metz, achieving the best performance in terms of three evaluation metrics, MSE, CI, and r^2 . On the KIBA dataset, TDGraphDTA slightly outperformed FusionDTA, GraphDTA, and MGraphDTA in terms of MSE, while its CI was slightly lower than that of MGraphDTA, still providing the best performance.

These findings underscored the effectiveness of TDGraphDTA as it outperformed other state-of-the-art DTA prediction models, demonstrating its superior performance.

The scatter plots and frequency histograms in Fig. 6 visualize the affinity prediction values and ground truth obtained from three datasets using a random splitting method. The distribution of ground truth and predicted values for TDGraphDTA is consistently similar across all three datasets, indicating that the predicted values effectively reflect the true affinity between drugs and targets.

4.3. Generalization performance and comparison of the model

Tables 3 to 5 present the experimental results using a random split setting, where the training and testing sets share the same drugs and targets. However, random splitting can lead to overly optimistic results due to potential information leakage (i.e., drug or target information) into the testing set. To further validate the effectiveness of multi-diffusion graph neural networks (MD-GNNs), in addition to the random split setting used in previous studies, MD-GNNs were also evaluated using three additional split schemes, [1,10,17]. Ablation experiments were conducted on three datasets based on whether the Transformer module was used and whether the Diffusion module was used. As shown in the results of ablation experiments, the application of Transformer and Diffusion can improve the prediction performance of DTA

Table 5

Performance comparison of TDGraphDTA and previous classic models on the Metz dataset. The best results are marked in bold.

Dataset	Category		Metz		
Model	Proteins	Compounds	MSE	CI(std)	r^2
RF	ECFP	PSC	0.351(0.002)	0.793(0.001)	0.565(0.001)
SVM	ECFP	PSC	0.361(0.001)	0.794(0.000)	0.590(0.001)
FNN	ECFP	PSC	0.316(0.001)	0.805(0.001)	0.660(0.003)
DeepDTA	CNN	CNN	0.286(0.001)	0.815(0.001)	0.678(0.003)
GraphDTA	CNN	GCN	0.282(0.007)	0.815(0.002)	0.679(0.008)
GraphDTA	CNN	GAT	0.323(0.003)	0.800(0.001)	0.625(0.010)
GraphDTA	CNN	GIN	0.313(0.002)	0.803(0.001)	0.632(0.001)
GraphDTA	CNN	GAT-GIN	0.282(0.011)	0.816(0.004)	0.681(0.026)
ML-DTI	CNN + Transformer	CNN + Transformer	0.322(0.009)	0.799(0.002)	0.623(0.015)
LSTM-DTI	LSTM	LSTM	0.325(0.011)	0.799(0.004)	0.609(0.022)
MGraphDTA	MCNN	MGNN	0.265(0.002)	0.822(0.001)	0.701(0.001)
TDGraphDTA	MCNN	With Transformer	0.269(0.005)	0.817(0.003)	0.710(0.003)
TDGraphDTA	MCNN	With Diffusion	0.274(0.003)	0.808(0.008)	0.706(0.006)
TDGraphDTA	MCNN	MD-GNN	0.264(0.003)	0.824(0.005)	0.711(0.003)

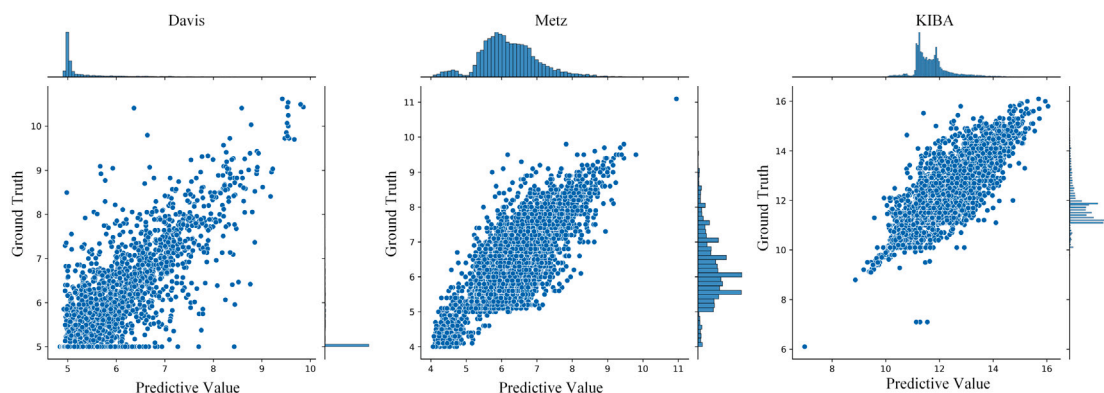


Fig. 6. The visualization of the scatter plots and frequency histograms of the predicted affinity values and ground truth on three datasets using random partitioning method.

to a certain extent, compared with other baseline models. Specifically, the Transformer mechanism has a greater improvement on CI and r^2 , while the Diffusion mechanism has a greater improvement on r^2 . It shows that the attention mechanism is effective in enriching feature information. The idea of diffusion is effective in optimizing the node information of molecules, enhancing the fitting degree of the model, and improving r^2 . The overall performance of the model is better than the individual performance of each module, and is also better than most baseline models in various performance indicators.

Given that DTA prediction models are commonly employed to discover missing drugs or targets within the training set, orphan splitting provides a more realistic and challenging evaluation scenario for the models. The adoption of cluster-based separation further safeguards against the inadvertent disclosure of compound structural information in the testing set. A comparative analysis between the proposed TDGraphDTA and GraphDTA was conducted, along with three traditional PCM models (RF, SVM, and FNN). To ensure fairness in the comparison, the source codes and reported hyperparameters provided by GraphDTA were used, substituting MD-GNN in TDGraphDTA with GCN, GAT, GAT-GCN, and MGNN. Model performance was assessed using a five-fold cross-validation strategy, where all methods shared the same training, validation, and testing sets, ensuring consistent experimental settings across the eight methods.

Fig. 7 presents the experimental results of eight methods using orphan-based and cluster-based splitting settings. Compared with the results obtained with the random splitting settings shown in Table 3 to Table 5, a significant decrease is obvious in model performance with the orphan-based and cluster-based splitting settings. As shown in Fig. 7(a) and (c), the TDGraphDTA model using orphan-compound splitting achieved MSE values of 0.581 ± 0.075 , 0.382 ± 0.021 , and

0.557 ± 0.046 on the Davis, KIBA, and Metz datasets, respectively. However, when employing cluster splitting, the corresponding MSE values were 0.657 ± 0.216 , 0.487 ± 0.097 , and 0.643 ± 0.075 . These results indicate that cluster-based splitting presents a greater challenge with the DTA prediction models compared with orphan-compound splitting, which aligns with the fact that cluster-based splitting prevents the leakage of structural information from compounds into the testing set. These findings highlight the ongoing challenge of improving the generalization ability of DTA models. Additionally, as illustrated in Fig. 7(a), MD-GNN significantly outperformed other methods ($p < 0.01$) on the Davis dataset using orphan-compound splitting. Similarly, although there were no statistically significant differences between these three methods ($p > 0.05$), MD-GNN, GAT, and RF significantly outperformed other methods ($p < 0.01$) on the KIBA dataset.

Furthermore, the SVM and FNN methods exhibited significant superiority over other methods on the Metz dataset ($p < 0.01$). Overall, the conventional PCM model showed impressive results, even surpassing graph-based methods employing orphan drug exclusion settings in both the KIBA and Metz datasets, as illustrated in Fig. 7(a). However, due to the noticeably smaller number of drugs in the Davis dataset compared with the KIBA and Metz datasets, as presented in Table 1, the generalization ability of models trained on a limited number of drugs cannot be guaranteed for unseen drugs. These results further validated the potential of MD-GNN to enhance the generalization capability of DTA models. From Fig. 7(b), it is evident that MD-GNN exhibited significant superiority over other models ($p < 0.05$) in the three datasets with orphan-target splitting configuration.

Using a clustering-based partitioning approach, MD-GNN exhibited significant superiority over other methods on the KIBA and Metz datasets, as shown in Fig. 7(c) ($p < 0.05$). Under the random split

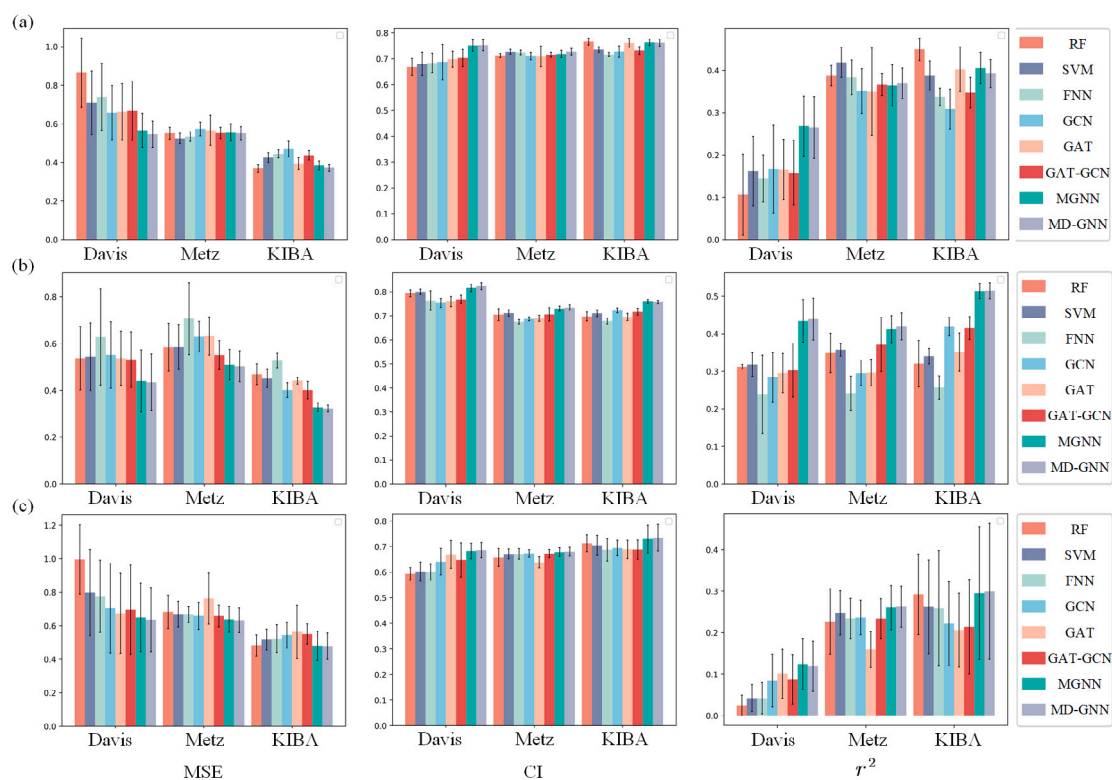


Fig. 7. Overview of performance metrics for datasets under different split methods. (a) Orphan drugs, (b) Orphan targets, and (c) Cluster-based partitioning settings. Compare MD-GNN with seven other models in terms of MSE, CI, and r^2 indices on the Davis, KIBA, and Metz datasets.

settings indicated in Tables 3 and 4, graph-based methods outperformed the traditional PCM model, while the advantage of graph-based methods was less pronounced under the orphan-based and clustering-based split settings shown in Fig. 7. Overall, the results showed the robustness of MD-GNN across different partitioning strategies and affirmed the vital role of capturing both local and non-local properties of molecules for accurate predictions with GNNs. This underscored the effective contributions made by the proposed multi-scale interaction block and diffusion-based Graph optimization block to the experimental outcomes.

4.4. Visualization of the model's attention on key molecular structures

Deep learning is often referred to as a black box model due to the challenge of attributing importance to specific features. The lack of interpretability hinders the application of deep learning methods, particularly in computer-aided drug discovery. To enhance the rationalization of graph-based models, Grad-CAM was used to visualize the atomic importance of molecules. In order to further understand the model's focus on structurally significant components, the ToxCast [46] biochemical dataset, that contains diverse toxicity property information [47], was used for visualization purposes. In the ToxCast dataset, the number of drugs (compounds), the number of targets (proteins), and the total number of drug–target affinity pairs are 7,657,335, and 530,605, respectively. Likewise, the training and validation sets were divided in a ratio of 8:2 based on the total number of drug–target affinities. The visualized heatmaps of molecular structures showed the ability of the model to capture toxicological structural information. Following the previously mentioned training approach, the model was trained on this dataset and Grad-CAM was used to visualize the importance of molecular structures in the testing set. These visualizations were conducive to checking whether the model can detect specific toxic substructures. The following two sets of experiments were conducted.

The model was trained in the ToxCast dataset. As shown in Fig. 8, the results of the attention layer were visualized, and the attention of the model to the toxic structure was presented on the two-dimensional molecular map. Attention visualization of different molecular two-dimensional structures of the same found toxic group structure [46,48] in short, long and symmetrical structures was used. A color map on the left side of the figure represents the level of attention (range from 0–1). In the ToxCast dataset, the model can find toxic structures in different positions and structures. It characterizes the ability of the model to capture the structure of interest, and shows that the model can achieve good performance through the feature fusion of two-dimensional graph learning and one-dimensional sequence learning through different methods. By querying in the ToxCast dataset, these compounds with toxic group structures all exhibit different levels of toxicity.

As shown in Fig. 9, six compounds containing different or the same toxic group structures in combination were randomly selected from ToxCast's random partition testing set, and a visualization of the model's attention to molecular structure is given. The left side of the figure shows the color map of the strength of the model attention (range from 0–1). The compounds in the figure have the CAS Number, which can be found in public biological databases such as ToxCast and PubChem. Among them, each compound molecule is divided into upper and lower parts, and the structure part marked in red in the two-dimensional molecular diagram above shows the different toxic structure combinations of the compound [49,50]. The two-dimensional molecular diagram below shows the model's attention to the structure of the compound. The figure shows that the model still has good capture ability in identifying different or identical toxicity alarm substructures in compounds with toxicity. But when toxic substructure coupling exists, it tends to focus on the most toxic structural combinations of toxic groups.

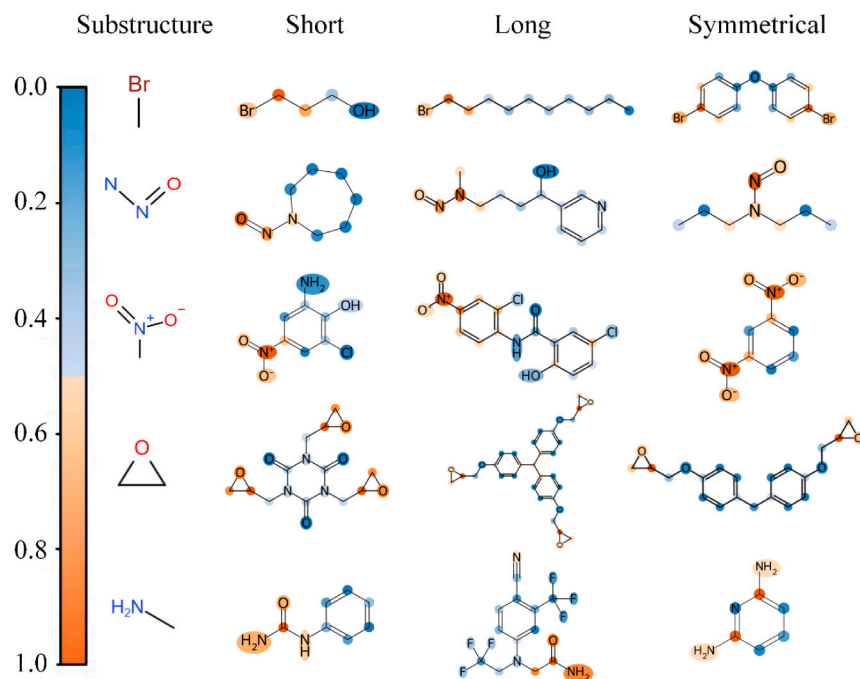


Fig. 8. Structural concern visualization of molecular toxic groups in ToxCast dataset. The left side marks the color map of the level of attention (range from 0–1). The above figure lists five types of toxic group structures, and shows the attention paid to toxic groups by the models under the short structure, long structure and symmetrical structure molecules. Among them, the five types of toxic groups are aliphatic, nitrosamines, specific aromatic nitro groups, epoxides, and aromatic amines.

4.5. Discussion

Existing DTA task methods have advantages in optimizing the drug screening process and effectively learning the structural characteristics of drug targets in public datasets, including the ability to process multi-scale information, the end-to-end learning framework to simplify feature engineering, etc. According to the performance effects of the above experiments, the combination of multi-scale information and graph optimization is effective for DTA tasks. Although TDGraphDTA shows significant improvement on the DTA prediction task, this study still has some limitations. First, the graph optimization method brings a large increase in the number of parameters, sacrificing time performance while obtaining better results. Trying to prune the model or perform other lightweight operations is an optimization direction. Second, many compounds containing structural alerts do not have specific toxicity profiles, and target-specific exploration in the form of drug–target combinations may be a worthwhile approach. The visualization of drug–target pair binding relationships is also one of the directions for future research optimization. It can reflect the model’s ability to capture drug–target pair interaction information and can also indicate the direction of model feature enhancement. Finally, this study does not integrate a larger biochemical dataset with drug–target pairing relationships. One of the future research undertakings would be to try larger biochemical datasets, such as validating the model on Therapeutics Data Commons (TDC) [51]. Specifically, for deep learning-based DTA tasks, the interpretability validation and performance counter-evidence of biochemical reactions between drug molecules and protein molecules are areas that require further research and efforts.

5. Conclusion

This paper introduces a novel deep neural network-based model called TDGraphDTA to predict DTA. The proposed model consists of

three main modules, the input module, molecular representation optimization learning module and the output module. TDGraphDTA takes drug compound graphs and protein sequences as inputs. The molecular representation optimization learning module consists of protein sequence feature representation learning, compound molecular graph feature representation learning, and multi-scale interaction block. The protein sequence feature representation learning uses multi-level cascaded CNN layers to combine receptive fields of different sizes and convolution channels of different depths to capture global and local sequence feature information. The compound molecular graph feature representation learning utilizes diffusion model-based graph optimization blocks to preserve relationship information among nodes in the graph. Therefore, it extracts additional molecular structural features, enriches structural representations, and provides interpretability of graphic representations. The multi-scale interaction block captures the relationships and contextual information between substructures of the molecule, enhances the important features associated with drug–target pairs, and improves the accuracy and reliability of predictions. The effectiveness of the TDGraphDTA model in DTA prediction is verified on benchmark datasets including Davis, Metz, and KIBA, using various data splitting methods as experimental settings. In addition, Grad-AAM was used to visualize the model’s attention to molecular structure, and ToxCast compounds with toxic structures were used for performance testing. It confirms that the TDGraphDTA model can show good attention to the substructure of intermolecular interactions through different ways of fused molecular representation learning, and can capture the substructure information of intermolecular interactions. The experimental results confirm that the proposed method is an effective and interpretable approach for DTA prediction, which can reduce time and costs in the drug discovery and screening processes. Although the proposed method performs well in the experimental results, there is still room for improvement in terms of model performance. For example, the performance of the modules used in the proposed method can be

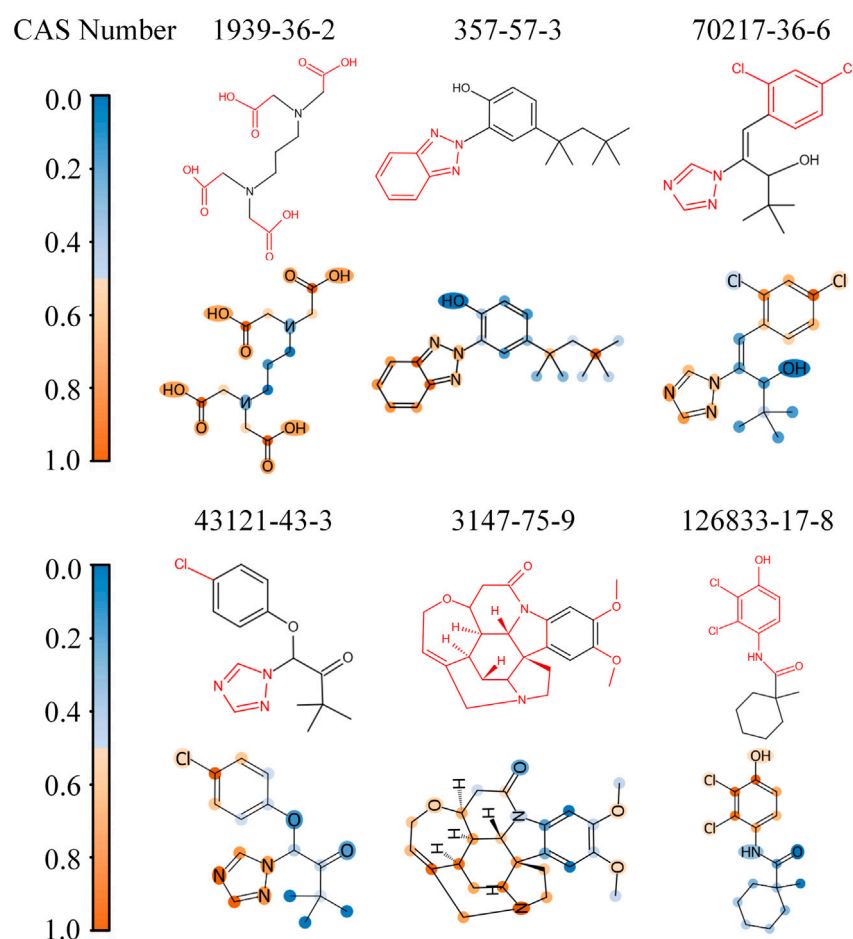


Fig. 9. Visualization of model attention to the presence of different combinations of toxic moiety structures in molecular structures of ToxCast dataset. The left side marks the color map of the level of attention (range from 0–1). The above figure lists the structures of toxic compounds that appear in six combinations of different toxic groups, and lists the CAS Number of the corresponding compounds. The related toxic groups can be found in ToxCast dataset, PubChem and other public biomedical dataset [46,48].

improved, by substituting different neural network modules, incorporating more molecular affinity feature information, and screening or testing more model hyperparameters.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used in this article are publicly available and can be accessed via the following links. The Davis dataset is available at <https://github.com/hkmztrk/DeepDTA/tree/master/data/davis>, the KIBA dataset is available at <https://github.com/hkmztrk/DeepDTA/tree/master/data/kiba>, the Metz dataset is available at https://github.com/simonfgy/PADME/tree/master/metz_data, and the ToxCast dataset is available at https://github.com/simonfgy/PADME/tree/master/full_toxcast.

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